Seoyul Oh

Ph.D. Candidate at the University of Illinois Urbana-Champaign (UIUC)

Thomas M. Siebel Center for Computer Science, 201 North Goodwin Avenue, Urbana, IL 61801

seoyuloh.github.io | seoyulo2@illinois.edu | in/seoyul-oh | @seoyuloh_012 | Google Scholar

Interests

My research focuses on AI for networked systems, including reinforcement learning for system optimization and AI agents for reliable infrastructure management. I also work on decentralized satellite networks.

Education

University of Illinois Urbana-Champaign, *IL, USA* — Sep 2023 – Present
Doctor of Philosophy, Computer Science
Advisors: Prof. Francis Y. Yan & Prof. Deepak Vasisht

Korea University, *Seoul, Korea* — Feb 2023
Master of Science, Electrical and Computer Engineering
Advisor: Prof. Sangheon Pack

Korea University, *Seoul, Korea* — Feb 2021
Bachelor of Science, Electrical Engineering

Employment

Apple | Connectivity, Cellular & Satellite Team *Cupertino, CA, USA* — May 2026 – Aug 2026
Software Engineering Intern
Developed network optimizations for reliable real-time video streaming over bandwidth-constrained and highly variable Direct-to-Cell LEO satellite links

Publications

[1] **Rediscovering Heuristics: A Litmus Test for RL in Networked Systems** — HotNets 2026
Seoyul Oh, Yikai Zheng, Eric Xiang Gao, Deepak Vasisht, Francis Y. Yan
ACM Workshop on Hot Topics in Networks

[2] **Exploiting Answer-Invariant Redundancies in Satellite Imagery for Efficient VLM Inference on Edge** — ML for Systems @ NeurIPS 2026
Ishani Janveja, Davis Zhang, **Seoyul Oh**, Deepak Vasisht
NeurIPS Workshop on Machine Learning for Systems (Spotlight)

[3] **EcoCell: Energy-aware Traffic Shaping for Cellular Radio Access Networks** — NINeS 2026
Zikun Liu, **Seoyul Oh**, Bill Tao, Anuj Kalia, Yaxiong Xie, Deepak Vasisht
New Ideas in Networked Systems

[4] **A Call for Decentralized Satellite Networks** — HotNets 2024
Seoyul Oh, Deepak Vasisht
ACM Workshop on Hot Topics in Networks

[5] **RIFT: Query-Aware Redundancy Sifting in Satellite Imagery for Efficient VLM Inference at the Edge**
Ishani Janveja, **Seoyul Oh**, Davis Zhang, Deepak Vasisht
Under Submission

[6] **OrbitShare: A Multi-Party Architecture for Decentralized LEO Satellite Networks**
Seoyul Oh, Bill Tao, Shreya Kannan, Deepak Vasisht
Under Submission

[7] **Towards Reliable Infrastructure Management with AI Agents**
Seoyul Oh, Nikhil Chandrasekar, Yiqi Mei, Deepak Vasisht, Francis Y. Yan
In Preparation

Research Experience

Illinois Networked Systems and AI (INSAI) Lab, University of Illinois Urbana-Champaign *IL, USA* Jan 2025 – Present
Research Assistant

Reinforcement Learning for Networked Systems
- Developed a framework for understanding why RL training succeeds or fails by testing whether simple policies can rediscover known heuristics and visualizing their learning behavior [1]

Agentic Infrastructure Management
- Developed graph-grounded approaches for reliable infrastructure diagnosis and incident mitigation [7]

Wonderful Wireless Lab, University of Illinois Urbana-Champaign *IL, USA* Sep 2023 – Present
Research Assistant

Federated Satellite Networks
- Designed decentralized architectures that enable multiple LEO satellite operators to share spare capacity for robust coverage without independently deploying large constellations [4] [6]

Energy-Efficient Edge Systems
- Developed energy-efficient systems through cellular traffic shaping [3] and redundancy-aware VLM inference for satellite imagery at the edge [2]

Mobile System, Security and Networking Lab, Purdue University *West Lafayette, IN, USA* Aug 2022 – Feb 2023
Visiting Graduate Researcher (Advisor: Prof. Chunyi Peng)
- Researched 5G mobility and connectivity through measurements using cellular-connected drones

Mobile Network and Communications Lab, Korea University *Seoul, South Korea* Mar 2021 – Jul 2022
Research Assistant
- Studied programmable data planes, network security, and resource allocation for federated edge-cloud systems

Honors and Awards

International Fellowship *American Association of University Women (AAUW) (25,000 USD)* Fall 2026

Korean Honor Scholarship *Ministry of Foreign Affairs, Korea (1,500 USD)* Fall 2025

KASF-KIA Scholarship *Korean American Scholarship Foundation (2,500 USD)* Fall 2025

International R&D Program Grant *Ministry of Science and ICT, Korea (12,580 USD)* Aug 2022 – Feb 2023

Teaching Experience

CS 538: Advanced Computer Networks, University of Illinois Urbana-Champaign *IL, USA* Spring, Fall 2025
Teaching Assistant
- Outstanding Teaching Assistant Award (Spring 2025)

Academic Service

Artifact Evaluation Committee ACM SIGCOMM '25, '26
Web Chair ACM SOSP PACMI Workshop '25

References

- Prof. Francis Y. Yan ..... *Assistant Professor in Computer Science at UIUC* [🌐]
- Prof. Deepak Vasisht ..... *Assistant Professor in Computer Science at UIUC* [🌐]
- Prof. Chunyi Peng ..... *Associate Professor in Computer Science at Purdue University* [🌐]
- Prof. Sangheon Pack ..... *Professor in Electrical Engineering at Korea University* [🌐]